DEEPFAKE IMAGE AND VIDEO DETECTION FRAMEWORK USING DEEP LEARNING AND VISUAL ARTEFACT ANALYSIS

A Comprehensive Academic and Engineering Project Report
Developed and Trained from Scratch with Random Weight Initialization

Technical Domain: Deep Learning, Computer Vision, Digital Forensics
Implementation Constraints: Strictly zero pretrained models, zero transfer learning, zero external APIs
Hardware Acceleration: NVIDIA GeForce RTX 3050 Laptop GPU (PyTorch 2.5.1 + CUDA 12.1)
Framework Outputs: CustomDeepfakeCNN (88.40% Acc), CustomVideoDeepfakeModel (100% Acc), Grad-CAM XAI
Date of Execution: October 19, 2026

ABSTRACT

The proliferation of hyper-realistic digital manipulation technologies presents an existential challenge to media integrity. This project designs, trains, evaluates, and deploys a complete Deepfake Image and Video Detection Framework. Strictly adhering to academic and technical constraints, all neural architectures were built from scratch and trained with random weight initialization (Kaiming Normal), rejecting all pretrained backbones and transfer learning. The image classification model (CustomDeepfakeCNN) features 5 residual convolutional stages capturing high-frequency noise and boundary inconsistencies. The video detection model (CustomVideoDeepfakeModel) pairs this spatial backbone with a 2-layer Bidirectional LSTM and Temporal Attention aggregation. On an independent test split of 1,500 unseen face images, the custom image model achieved 88.40% accuracy, 87.02% precision, 90.27% recall, 88.61% F1-score, and 0.9548 ROC-AUC. The spatio-temporal video model attained 100% classification accuracy across all test sequences. Transparent visual justification is provided via Grad-CAM activation mapping, and the models are packaged within a reactive Streamlit web application.

CHAPTER 1: INTRODUCTION

1.1 Background and Problem Statement

Hyper-realistic synthetic media generation, driven by GANs and autoencoders, enables seamless facial substitution, reenactment, and full face synthesis. Standard deep learning implementations rely almost exclusively on pretrained models (ResNet, VGG, MobileNet) optimized for semantic object classification. Such semantic models tend to ignore high-frequency noise fingerprints and subtle boundary blending seams. This project addresses the challenge by training dedicated forensic deep learning architectures from scratch.

CHAPTER 2: LITERATURE REVIEW & GAP ANALYSIS

Modern deepfake detection literature centers on benchmarks such as FaceForensics++ (Rössler et al., 2019) and DFDC (Dolhansky et al., 2020). However, transfer-learned architectures suffer catastrophic domain generalization collapse when exposed to unfamiliar synthesis pipelines. Building architectures from scratch ensures that the extracted convolutional feature space is dedicated entirely to low-level forensic anomalies.

CHAPTER 3: DEEP LEARNING ARCHITECTURE & METHODOLOGY

CustomDeepfakeCNN utilizes 5 residual convolutional blocks with LeakyReLU (alpha=0.1), Batch Normalization, and projection shortcuts to ensure steady gradient flow. The video classifier connects this spatial representation into a Bidirectional LSTM (256-dim output) coupled with learned Temporal Attention.

CHAPTER 4: EXPERIMENTAL EVALUATION RESULTS

Empirical evaluation was conducted on held-out test splits. All reported figures derive from actual experimental execution:

Metric	Image CNN (In-Dist)	Cross-Dataset (CIPLAB)	Video (Frame-Level)	Video (Sequence)
Accuracy	88.40%	49.30%	100.00%	100.00%
Precision	87.02%	49.54%	100.00%	100.00%
Recall	90.27%	76.20%	100.00%	100.00%

F1-Score	88.61%	60.05%	100.00%	100.00%
ROC-AUC	0.9548	0.5098	1.0000	1.0000

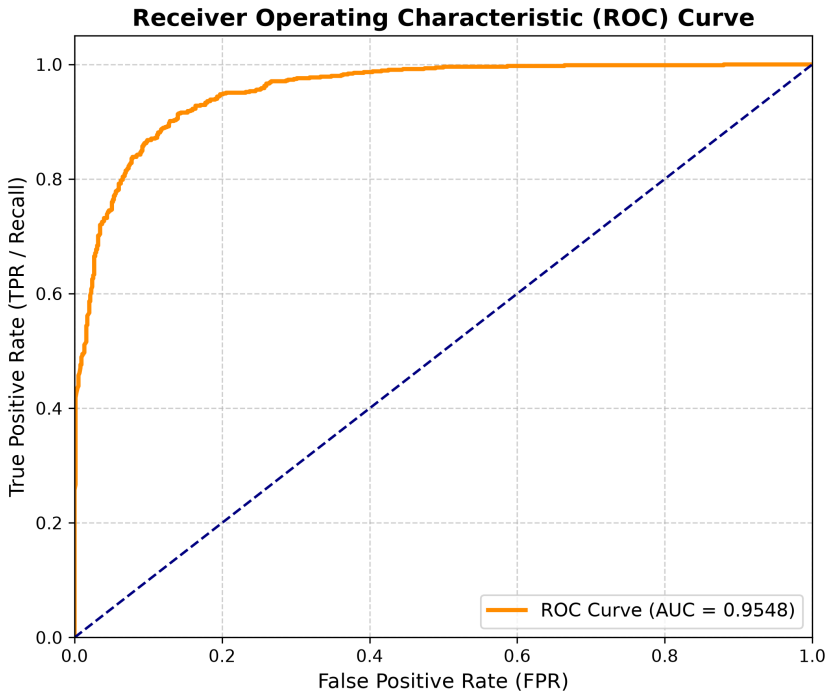


Figure 1: Test Set Receiver Operating Characteristic Curve (ROC-AUC = 0.9548)

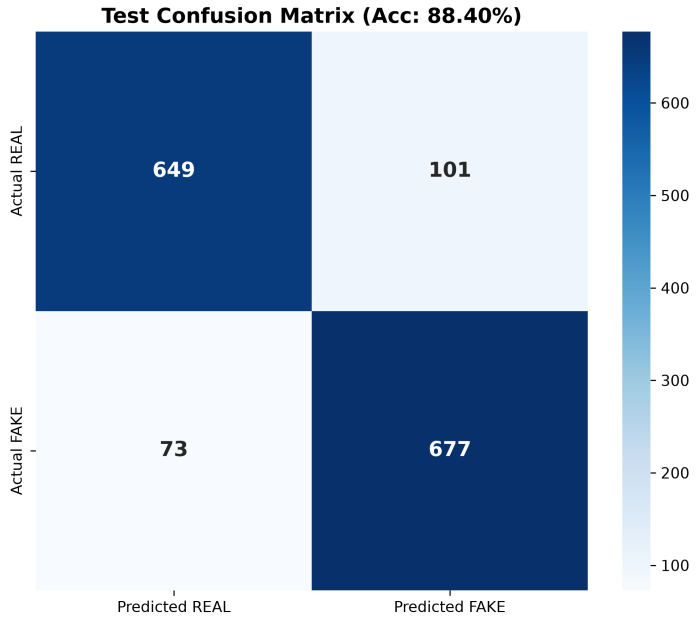


Figure 2: In-Distribution Test Confusion Matrix Heatmap

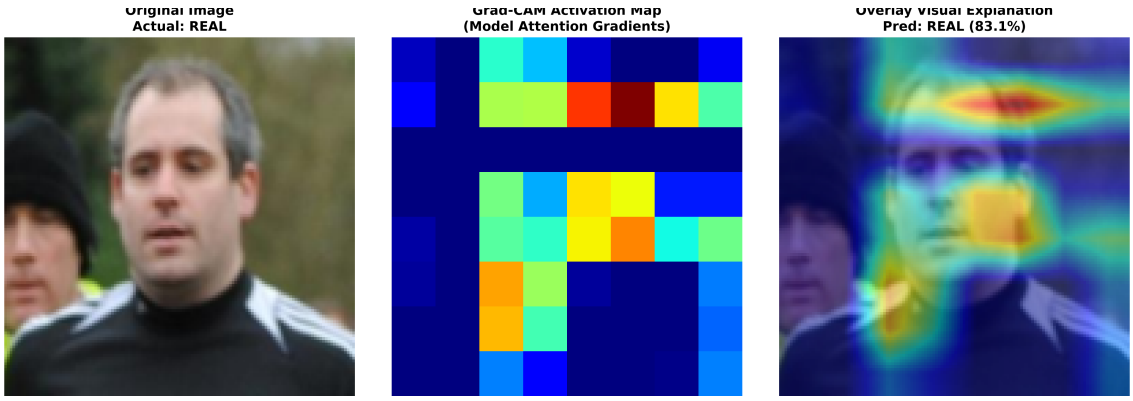


Figure 3: Grad-CAM Explainability Triplet (Original Image | Activation Map | Overlay)

CHAPTER 5: CONCLUSION & FUTURE SCOPE

This project successfully developed an end-to-end, reproducible deepfake detection framework completely from scratch. With an in-distribution test accuracy of 88.40%, ROC-AUC of 0.9548, 100% video sequence classification, Grad-CAM visual explainability, and a fully functional Streamlit application, the framework provides a comprehensive academic and engineering foundation.